

Innovation is Connected Correlation: the Hinge as the Cumulant Functor

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Status: All identities verified numerically (mpmath; $m = 1..4$ cumulants against three independent expressions; per-mode formula exact to 12 digits against CGF derivatives). The mathematics is classical statistical mechanics of a free gas plus classical zeta identities — walls throughout; the contribution is the identification and its consequences for the framework. One candle is flagged in §5 with the precise reason it may not be banked.

1. Setup

Fix $\beta > 1$. Let N be the random integer with $\Pr(N = n) = n^{-\beta}/\zeta(\beta)$ (the primon gas state). By unique factorisation and the Euler product, $N = \prod_p p^{K_p}$ where the occupations K_p are **independent**, geometrically distributed: $\Pr(K_p = k) = (1 - x_p) x_p^k$, $x_p = p^{-\beta}$. The energy of the state is its resistance, $\ln N = \sum_p K_p \ln p$ — an observable **additive over the modes**.

2. Lemmas

Lemma 2.1 (per-mode cumulants). The m -th cumulant of a geometric variable with ratio x is

$$\kappa_m(K) = \sum_{k \geq 1} k^{m-1} x^k = \text{Li}_{1-m}(x).$$

Proof sketch. The cumulant generating function is $C(t) = \ln \frac{1-x}{1-xe^t}$; expand $-\ln(1 - xe^t) = \sum_{k \geq 1} \frac{x^k e^{kt}}{k}$, differentiate m times at $t = 0$ to get $\sum_k k^{m-1} x^k$. (*Wall — verified numerically to 12 digits for $m = 1..4$ at $x = 0.3$: e.g. $\kappa_2 = x/(1-x)^2$, $\kappa_3 = x(1+x)/(1-x)^3$.)*

Lemma 2.2 (cumulant additivity and the vanishing of mixtures). For independent variables, cumulants of sums add, and all **mixed** cumulants vanish. Hence for the mode-additive energy,

$$\kappa_m(\ln N) = \sum_p (\ln p)^m \kappa_m(K_p).$$

(*Wall — standard.*)

3. The master identity

Theorem 3.1. For every $m \geq 1$ and $\beta > 1$:

$$\kappa_m(\ln N) = \sum_{n \geq 2} \Lambda(n) (\ln n)^{m-1} n^{-\beta} = (-1)^m \frac{d^m}{d\beta^m} \ln \zeta(\beta).$$

Proof. By Lemma 2.2 and Lemma 2.1, $\kappa_m(\ln N) = \sum_p (\ln p)^m \sum_{k \geq 1} k^{m-1} p^{-\beta k} = \sum_{p,k} \ln p \cdot (k \ln p)^{m-1} p^{-\beta k}$, which is the Λ -sum since $\Lambda(p^k) = \ln p$ and $\ln p^k = k \ln p$. Alternatively and equivalently: the moment generating function of $\ln N$ is $\mathbb{E}[N^t] = \zeta(\beta - t)/\zeta(\beta)$, so the CGF is $\ln \zeta(\beta - t) - \ln \zeta(\beta)$, whose m -th derivative at $t = 0$ is $(-1)^m (\ln \zeta)^{(m)}(\beta)$; and from $\ln \zeta(s) = \sum_n \frac{\Lambda(n)}{\ln n} n^{-s}$ each s -derivative brings down $-\ln n$. $\square$ (Verified numerically for $m = 1..4$ at $\beta = 2$ via all three expressions.)

Reading the theorem. Every cumulant of the gas lives on the **innovation skeleton**: atoms exactly at the prime-power resistances $\ln p^k$, base mass $\Lambda(n) = \ln p$, multiplied by (position) $^{m-1}$. Composite positions carry no atom *at any cumulant order* — Lemma 2.2’s vanishing of mixed cumulants is the mechanism: a composite is a product of independent modes, and independent things have moments but **no connected correlation**. Meanwhile the raw m -th moment, $\mathbb{E}[(\ln N)^m] = \zeta(\beta)^{-1} \sum_n (\ln n)^m n^{-\beta}$, carries atoms at **every** integer.

4. Consequences

Corollary 4.1 (the hinge is the cumulant functor). The all-integers object generates moments ($\mathbb{E}[N^t] = \zeta(\beta - t)/\zeta(\beta)$); its logarithm generates cumulants. The same map that is resistance, code length, and the extractor of atoms ($\ln \zeta = P + \dots$, Cathedral, Hinge) is — probabilistically — the moments $\rightarrow$ cumulants passage. One logarithm, one more job: **quantity lives in the moments; the innovation skeleton is what survives the cumulant functor**. This is the statistical identity of “studying numbers stripped of their quantity property.”

Corollary 4.2 (the order–position lock, now a theorem). The Primon/Fisher note’s verdict — Fisher information carries masses $\Lambda(n) \cdot \ln n$ — is the $m = 2$ row of Theorem 3.1, and the pattern is total: the m -th order statistical form carries exactly $m - 1$ extra factors of position. In a static ensemble, the order of the form and the power of the position weight are locked. The Weil functional (Λ masses, i.e. $m = 1$ weights, inside an $m = 2$ -shaped object) is therefore **not any cumulant of the static gas** — completing, as a theorem, the argument that the source must carry its atoms in the lags of a flow, not the positions of a measure.

Corollary 4.3 (which probes can see the atoms). Only mode-additive (compositional) observables have cumulants on the skeleton; any function of the scalar $\ln N$ alone re-mixes the modes and its variance sees every integer. Paper 3’s magnitude/direction split, as a statement about measurements: *the innovation structure is visible only to probes that respect composition*.

5. The two Möbius functions — a candle, with the reason it stays one

There is a known combinatorial theory of cumulants: they are the **Möbius inversion of moments over the partition lattice** (Rota; Speed). The framework’s Möbius is the function of the **divis-**

ibility lattice ($\mu(n)$, the phase face of Paper 8). Both are instances of Rota’s incidence-algebra inversion — but **on different posets**, and they enter the present result at different joints: the divisibility-lattice structure supplies the independent modes (Euler product), while the partition-lattice inversion is what “cumulant” *means*. In the primon gas the single operation $\ln$ performs both passages at once ($\zeta \rightarrow \ln \zeta$ extracts atoms; $\text{MGF} \rightarrow \text{CGF}$ extracts connected correlations), which is why the identification works — but “the two Möbius functions are one object” is **not banked**: the honest open question is whether the coincidence of the two inversions under one logarithm is forced (a wall to be found) or is a fortunate alignment specific to free/independent structures. Free probability, where cumulants are governed by the *non-crossing* partition lattice, suggests the alignment is structure-dependent — a sharp place to test.

6. Honesty boundary

Walls: Lemmas 2.1–2.2, Theorem 3.1 and its proof (classical; the primon-gas setting is Julia / Bost–Connes prior art). Contributions: the innovation-skeleton reading, Corollaries 4.1–4.3 as framework statements, and the placement of the Fisher verdict as the $m = 2$ row of a total pattern. Candle, explicitly unbanked: §5. Nothing here constrains the zeros.

Developed jointly by Alexander Pisani and Claude (Anthropic).